CUCUMBERCARE AI — LEAF HEALTH ASSESSMENT REPORT

Automated Agronomic Screening Assessment • Generated October 29, 2026 at 23:46 UTC • Model: ConvNeXt Small (8 Classes)

SECTION A: PRIMARY DISEASE OR HEALTH CLASSIFICATION (IMAGE CLASSIFIER OUTPUT)

Predicted Condition:	**Nutrient Deficiency**
Scientific Designation:	*Abiotic Physiological Condition*
Model Confidence:	**53.1%** (Moderate Confidence)
Diagnostic Engine:	PyTorch ConvNeXt (ixrbhii/cucumber-disease-convnext) + Static_botanical Advisor

Pathology Overview: An abiotic, non-infectious physiological disorder resulting from soil nutrient depletion, pH lockout, or root zone waterlogging that prevents adequate mineral translocation.

Differential Identification: Symmetrical patterns across leaves (such as uniform interveinal chlorosis or bilateral marginal yellowing) with clean undersides lacking fungal spores or bacterial ooze.

Primary Visible Symptoms:
- Interveinal chlorosis (yellow tissue between dark green veins) on young upper leaves (Iron or Manganese deficiency)
- Uniform yellowing beginning on oldest lower foliage (Nitrogen deficiency)
- Marginal scorching or yellowing along leaf perimeters (Potassium deficiency)

Recommended Immediate Field Actions:
1. Test root zone soil pH and Electrical Conductivity (EC) immediately before applying treatments.
2. Apply a foliar micronutrient spray containing chelated Iron, Magnesium, and Zinc for quick correction.
3. Ensure proper root zone aeration, verify drainage, and avoid overwatering.

Long-Term Prevention & Cultural Practices:
- Perform regular soil nutrient testing before each planting cycle.
- Use balanced, buffered water-soluble fertilizers tailored to cucumber growth stages.
- Incorporate well-decomposed organic compost to improve cation exchange capacity.

Treatment Caution: Do not guess fertilizer rates or apply excessive nitrogen, which can burn roots and induce secondary calcium deficiency in developing fruit. **When to Seek Expert Consultation:** Consult a certified crop advisor or soil laboratory if symptoms worsen after nutrient correction, as root nematodes or vascular root rot may be impairing nutrient uptake.

SECTION B: NUTRITIONAL AND PHYSIOLOGICAL ASSESSMENT

The image classifier identified Nutrient Deficiency as the primary condition. Below is the targeted assessment of possible nutritional and root-zone causes.

Nutrient Element	Visual Foliar Symptom Pattern	Differential from Infectious Disease
Nitrogen (N)	Uniform pale-green to chlorotic yellowing appearing first on oldest, lower leaves, moving inward across the entire foliage	Yellowing is uniform across the leaf surface, unlike Downy Mildew's ang
Potassium (K)	Marginal chlorosis advancing to crisp necrosis and bronzing along outer leaf edges, often curling upward	Symptoms are physiological, not caused by pathogens, so no fungal spores
Magnesium (Mg)	Distinct interveinal chlorosis (yellowing between primary veins while main veins remain dark green), older leaves affected first	Major veins remain dark green, unlike Powdery Mildew which coats the v
Iron (Fe)	Severe, vivid interveinal chlorosis appearing strictly on the youngest expanding tip leaves; in severe cases leaves turn entirely white	Restricted to new growth; lower mature leaves remain healthy, unlike nitrogen and m

Model Probability Distribution (Top Candidates)

Rank	Candidate Condition	Model Probability	Classification Status
#1	Nutrient Deficiency	53.1%	Supported Class

#2	Healthy Leaves	22.8%	Primary Class
#3	Angular Leaf Spot	7.1%	Supported Class

Responsible AI Notice & Limitation of Liability: *Educational Screening Notice: This analysis is generated by an automated AI image classifier. Visual symptoms can mimic other physiological stresses. Verify with local agricultural extension or professional crop advisors prior to applying chemical treatments.*